Alivelli Ashwanth

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PROFESSIONAL SUMMARY

AI Engineer with 1.5+ years of experience designing, building and deploying production Generative AI applications for cybersecurity and regulated insurance workflows. Builds Retrieval-Augmented Generation (RAG) systems, AI agents and compliance-aware LLM pipelines with Python, LangChain, LangGraph and FastAPI, backed by vector search on FAISS, ChromaDB, Pinecone and MongoDB Atlas. Brings a security-first approach to AI through prompt injection defense, guardrails, automated red-teaming and LLM evaluation. Experienced with OpenAI, Google Gemini and Anthropic Claude APIs, and with fine-tuning open-source LLMs using LoRA and QLoRA.

EXPERIENCE

Associate AI Engineer

Cygne Noir Cyber — November 2025 - Present

Insurance Rate Filing Automation - California P&C Insurance Client

- Built a **RAG system** using the Google Gemini API and MongoDB to automate California Department of Insurance Prior Approval rate filings for Property and Casualty lines under Proposition 103.
- Engineered a **regulatory authority hierarchy** and compliance guardrails so generated outputs follow state rules and cite the governing regulation.
- Implemented field-by-field analysis of actuarial filing documents covering loss development factors (LDF), defense and cost containment expenses (DCCE), MPEP, credibility complements and SERFF submission requirements.

Nudr - Dating App Recommendation Engine

- Developed an embedding-based **recommendation engine** for Nudr that represents each user's profile, interests and stated preferences as vectors to score compatibility between users.
- Implemented two-stage matching: fast candidate retrieval with **Pinecone vector search**, followed by re-ranking on location, recent activity and similarity scores.
- Added a **behavioral feedback loop** that learns from likes, skips and messages to refine recommendations over time.

IRIS - AI-Driven Incident Response Platform

- Built core modules of IRIS, an AI-driven incident response platform that uses LLMs to analyze security alerts, correlate related events and generate guided response recommendations.
- Developed backend services and **LLM pipelines** in Python and FastAPI, using structured outputs to produce consistent, analyst-ready incident reports.

Echosim - Adversary Emulation and Breach & Attack Simulation (BAS) Engine

- Developed adversary emulation capabilities for Echosim, a BAS engine that simulates real-world attack techniques mapped to the **MITRE ATT&CK** framework to validate security controls.
- Applied LLMs to generate attack scenarios and produce readable simulation reports for security teams.

LLM Agent Security Gateway with Automated Red-Teaming (Internal POC)

- Built a **guardrail gateway** for LLM agents that detects prompt injection, jailbreaks, PII leakage and unsafe tool calls in real time.
- Developed an **automated red-teaming agent** in LangGraph that generates direct and indirect prompt injection attacks mapped to the OWASP Top 10 for LLM Applications.
- Benchmarked attack success rate, false-positive rate and latency overhead with and without the gateway, and shipped it as a Dockerized FastAPI service with an OpenAI-compatible API.

AI Engineer Trainee

Cygne Noir Cyber — March 2025 - October 2025

Internal RAG Chatbot

- Built a **RAG-based chatbot** that answers employee questions over internal company documentation with source-cited responses, replacing manual document search.
- Implemented the ingestion pipeline with document parsing, chunking, embeddings and vector search, and tuned retrieval and prompts to keep answers grounded.

SIEM Log Generation

- Built synthetic log generators for SIEM platforms using **few-shot and chain-of-thought prompting**, producing realistic firewall, IDS and authentication events for detection testing.

LLM Threat Intelligence Tools

- Developed LLM-based threat intelligence tools that turn unstructured threat reports into structured, analyst-ready intelligence for security use cases.

PROJECTS

Fine-Tuned LLM for Detection Rule Generation

Personal Project · GitHub

- Fine-tuned an open-source 7B LLM with **QLoRA** using Hugging Face Transformers and PEFT on synthetic attack log and Sigma rule pairs to generate SIEM detection rules from attack descriptions.
- Built an **execution-based evaluation harness** that runs generated Sigma rules against held-out attack and benign logs to measure detection rate and false positives.
- Compared the fine-tuned model against a frontier API model on accuracy, cost and latency, and served it with vLLM behind FastAPI.

CNC Machine Failure Prediction

Final Year Project, MLR Institute of Technology · Jan 2024 - Apr 2024

- Collected real-time vibration, temperature, spindle speed and current data from sensors on the college's CNC machines during live machining operations.
- Designed a preprocessing pipeline that cleaned and labeled the live sensor streams, then derived machine health indicators such as rolling averages, vibration peaks and temperature trends over **sliding time windows**.
- Trained an **XGBoost classifier** that flags abnormal machine behavior and forecasts failures in real time, reaching 95% accuracy, with precision, recall and F1-score tracked to limit missed failures.

EDUCATION

Bachelor of Technology - Computer Science and Engineering (Data Science)

MLR Institute of Technology, Hyderabad · 2021 - 2024 · CGPA: 8.0/10

SKILLS

Generative AI & LLMs: Large Language Models (LLMs), Generative AI, Retrieval-Augmented Generation (RAG), Agentic AI, AI Agents, Multi-Agent Systems, Prompt Engineering, Tool/Function Calling, Model Context Protocol (MCP), Structured Outputs, Fine-Tuning (LoRA, QLoRA), LLM Evaluation, Guardrails, Hallucination Mitigation

LLM Frameworks & APIs: LangChain, LangGraph, LlamaIndex, Hugging Face Transformers, PEFT, vLLM, Ollama, LangSmith, OpenAI API, Google Gemini API, Anthropic Claude API

Vector Databases & Retrieval: FAISS, ChromaDB, Pinecone, MongoDB Atlas Vector Search, Embeddings, Semantic Search, Hybrid Search (BM25), Re-ranking

Machine Learning & NLP: PyTorch, scikit-learn, XGBoost, Recommendation Systems, NLP, Transformers, Deep Learning, pandas, NumPy

Programming & Databases: Python, SQL, Java, MongoDB, PostgreSQL, MySQL, Redis

Backend, MLOps & Cloud: FastAPI, REST APIs, Pydantic, Docker, Git, GitHub Actions (CI/CD), AWS, Linux

CERTIFICATIONS & ACHIEVEMENTS

- TCS CodeVita Season 10: Global Rank 4533 (2022)
- HackerRank: Gold badges in Python and Java, Silver badge in SQL
- Smart Interviews: Smart Coder certification in Data Structures and Algorithms
- NPTEL: Data Analytics with Python (2023)
- IBM: Python for Data Science (2021)